

STAT/MA 41600

In-Class Problem Set #1: August 26, 2026

1. Flip 100 coins in a row. Each element in the sample space is an ordered list of 100 flips.
 - 1a. How many outcomes are in the sample space altogether?
 - 1b. How many outcomes have exactly 50 heads and 50 tails?
2. Roll a green 6-sided die and a red 6-sided die.
 - 2a. How many outcomes have a sum between 4 and 8 (inclusive)?
 - 2b. If X is the value on the green die and Y is the value on the red die, how many outcomes have $|X - Y| \leq 1$? In other words, in how many outcomes are the two dice values the same or differing by 1?
3. Consider a randomly chosen bear family, consisting of 3 bears, namely, a daddy bear, a momma bear, and a baby bear. Each bear has six possible colors, namely, red, orange, yellow, green, blue, purple. If we pick such a bear family randomly, the sample space contains all ordered 3-tuples whose elements are r, o, y, g, b, p .
 - 3a. How many bear families are there? I.e., how many outcomes are in the sample space?
 - 3b. How many bear families do not contain any red bears? (De Morgan's First Law: If A_1 is the event that the daddy bear is red, and A_2 is the event that the momma bear is red, and A_3 is the event that the baby bear is red, then we are asking how many outcomes are in the event $(A_1 \cup A_2 \cup A_3)^c$).
 - 3c. We say that a bear family is "primary" if it has a red, a blue, and a yellow bear (in any order). How many bear families are *not* primary? (De Morgan's Second Law: If B_1 is the event that the bear family has a red bear, and B_2 is the event that the bear family has a blue bear, and B_3 is the event that the bear family has a yellow bear, then we are asking how many outcomes are in the event $(B_1 \cap B_2 \cap B_3)^c$).
4. During a 5-day week (Monday to Friday), each day is sunny, cloudy, or rainy. If we pick a week randomly, the sample space contains all ordered 5-tuples whose elements are s, c , or r . One outcome in the sample space is (s, s, c, r, s) . Another outcome is (c, r, c, s, s) .
 - 4a. How many weekly weather forecasts are possible for the 5-day week? (In other words, how many outcomes are in the sample space?)
 - 4b. How many weekly weather forecasts have 0 sunny days?
 - 4c. How many weekly weather forecasts have 1 sunny day?
 - 4d. How many weekly weather forecasts have 2 sunny days?
 - 4e. How many weekly weather forecasts have 3 sunny days?
 - 4f. How many weekly weather forecasts have 4 sunny days?
 - 4g. How many weekly weather forecasts have 5 sunny days?

(Hint: Make sure that your answer to **4a** is the same as the sum of **4b** through **4g**.)
5. Calculus review. Find a closed form expression for each of these:
 - 5a. $\int_0^\infty 3e^{-3x} dx$
 - 5b. $\int_0^\infty (x)(3e^{-3x}) dx$
 - 5c. $\sum_{x=0}^\infty \frac{3^x e^{-3}}{x!}$
 - 5d. $\sum_{x=2}^\infty \frac{3^x e^{-3}}{x!}$